Department
of Mathematics and
Physics, School of arts and
Science, Qatar University.

Unit Name:	Mathematics for Engineering	Instructor:	Dr. Mohamed Mabrok
Section:	L52	Location:	
Exam :	Midterm one	Time and Date:	80 minutes. October, 10, 2022

Student Name:	
Student ID:	

- No credit will be given to answers for non-requested questions.
- Only non-plotting calculators are permitted. The use of the cell phones is absolutely prohibited.
- Shut off your cell phone completely.
- Make sure that you are feeling comfortable before entering the class for the test.
- Full mark will be given for complete answers for all questions.
- Cheating is an academic violation according to Qatar University rules and regulations, and in some cases, it may result in final dismissal from the University.

Question Number	Student Marks	Maximum Marks
1		**6**
2		**6**
3		**5**
4		**6**
5		**6**
6		**5**
Total / (Max)		**30/34**

Questions:

1. Solve the following differential equation the equation:

$$\csc(x)\,\frac{\mathrm{d}y}{\mathrm{d}x}-\frac{1}{x}\csc(x)\,y=y^3.$$

6 Mark

2 Solve the following differential equation:

$$(2xy^3-2x^3y^3-4xy^2+2x)dx+(3x^2y^2+4y)dy=0$$

6 Mark

3 Find the general solution of:

$$y''+9y=\csc 3x$$

5 Mark

4 Solve the following differential equation:

$$(t+1)^2y''+3(t+1)y'+2y=0$$

6 Mark

5 Find the general solution of the following differential equation if $y_1 = e^x$ is a solution of the homogenous part of the following DE.

$$xy'' - y' + (1-x)y = 4x^2 \mathrm{e}^x$$

6 Mark

6 Match each differential equation with it's particular solution form.

Number	Y_p	Answer	DE
1)	Ate^{3t}		$y''+6y'+9y=4e^{-3t}$
2)	At^2+Bt+C		$y''-9y'+20y=3t^2+5t-2$
3)	$A\cos(3t)+B\sin(3t)$		$y''+6y'+9y=4\cos(3t)$
4)	Ae^{-4t}		$y''-9y'+20y=e^{4t}$
5)	At^2e^{-4t}		$y''+6y'+9y=te^{4t}$
6)	$(At+B)e^{3t}$		$y''-9y'+20y=e^{-3t}$

5 Mark

Algebra

- $\sin\theta = \frac{1}{\csc\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$
- $\cot\theta = \frac{1}{\tan\theta}$
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$
- $\cos^2(x) = 1 - \sin^2(x)$
- $\sin^2(x) = 1 - \cos^2(x)$

Differentiation

- $\frac{\mathrm{d}}{\mathrm{d}x} c = 0$
- $\frac{\mathrm{d}}{\mathrm{d}x} x^n = nx^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} (ax \pm b)^n = a\,n(ax \pm b)^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|ax \pm b| = \frac{a}{ax \pm b}$
- $\frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = ae^{ax}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\cos(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\sin(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} f(x)g(x) = f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x}$ *Product rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} \frac{f(x)}{g(x)} = \frac{g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x} - f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x}}{(g(x))^2}$ *Quotient rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$ *Chain rule.*

Integration

- $\int a\,dx = ax + c$
- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad n \neq -1$
- $\int f(x)^n f'(x)\,dx = \frac{f(x)^{n+1}}{n+1} + c \quad n \neq -1$
- $\int \frac{a}{x}\,dx = a\ln|x| + c$
- $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + c$
- $\int e^{ax}\,dx = \frac{1}{a}e^{ax} + c$
- $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$
- $\int \sin(ax)\,dx = \frac{-1}{a}\cos(ax) + c$
- $\int \cos(ax)\,dx = \frac{1}{a}\sin(ax) + c$
- $\int \sec^2(ax)\,dx = \frac{1}{a}\tan(ax) + c$
- $\int \csc^2(ax)\,dx = -\frac{1}{a}\cot(ax) + c$
- $\int \tan(ax)\,dx = -\frac{1}{a}\ln(\cos(ax)) + c$
- $\int \sec(ax)\tan(ax)\,dx = \frac{1}{a}\sec(ax) + c$
- $\int \csc(ax)\cot(ax)\,dx = -\frac{1}{a}\csc(ax) + c$
- $\int \sec(ax)dx = \frac{1}{a} \ln(\sec(ax) + \tan(ax)) + c$
- $\int \csc(ax)\ dx = \frac{-1}{a} \ln(\csc(ax) - \cot(ax)) + c$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + C$